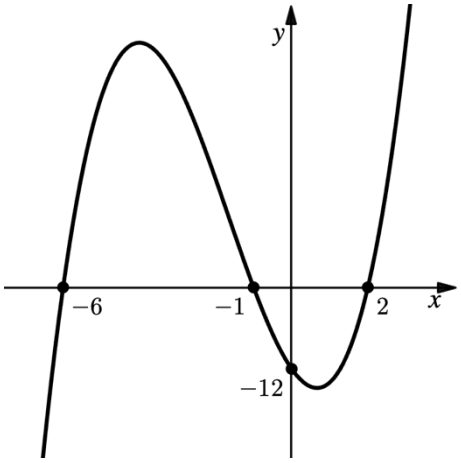
1) a)



1) b) i)

$\{x: x \in \mathbb{R}, -6 < x < -1\} \cup \{x: x \in \mathbb{R}, x > 2\}$

1) b) ii)

$\{x: x \in \mathbb{R}, x \leq -6\} \cup \{x: x \in \mathbb{R}, -1 \leq x \leq 2\}$

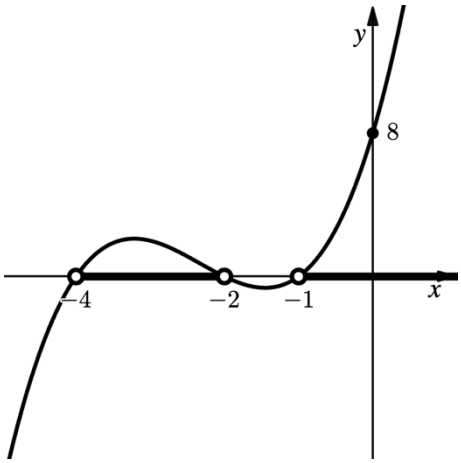
1) c) i)

$x \in (-\infty, -6) \cup (-1, 2)$

1) c) ii)

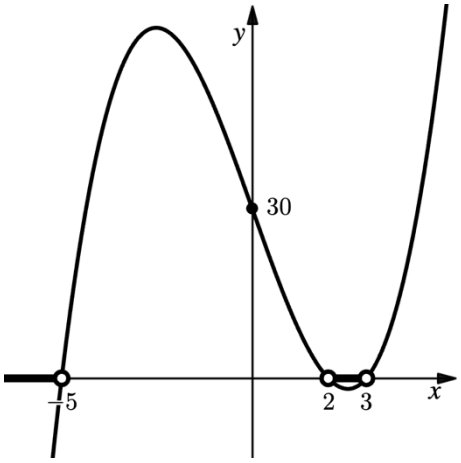
$x \in [-6, -1] \cup [2, \infty)$

2) a)



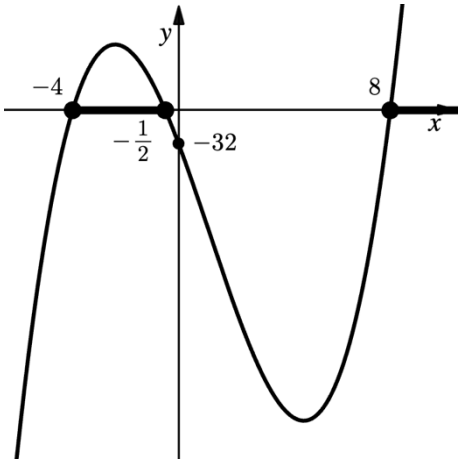
$\{x: x \in \mathbb{R}, -4 < x < -2\} \cup \{x: x \in \mathbb{R}, x > -1\}$

2) b)



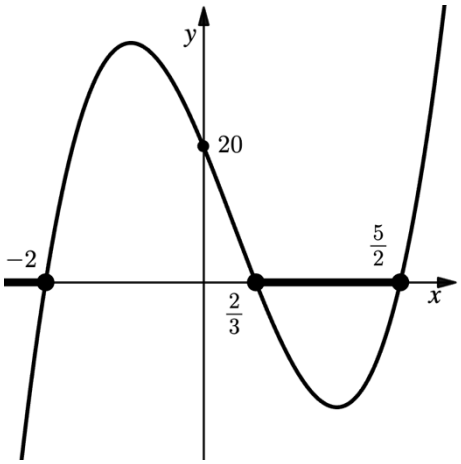
$\{x: x \in \mathbb{R}, x < -5\} \cup \{x: x \in \mathbb{R}, 2 < x < 3\}$

2) c)



$\left\{x: x \in \mathbb{R}, -4 \leq x \leq -\frac{1}{2}\right\} \cup \{x: x \in \mathbb{R}, x \geq 8\}$

2) d)

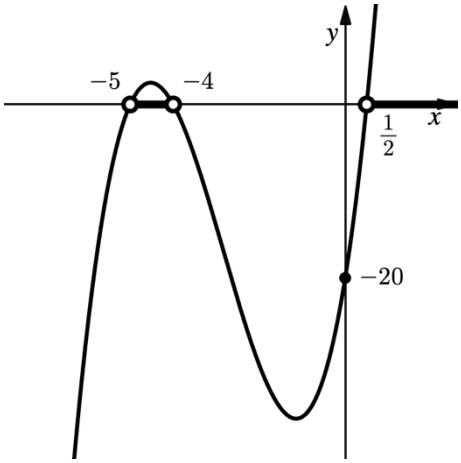


$\{x: x \in \mathbb{R}, x \leq -2\} \cup \left\{x: x \in \mathbb{R}, \frac{2}{3} \leq x \leq \frac{5}{2}\right\}$

2) e)

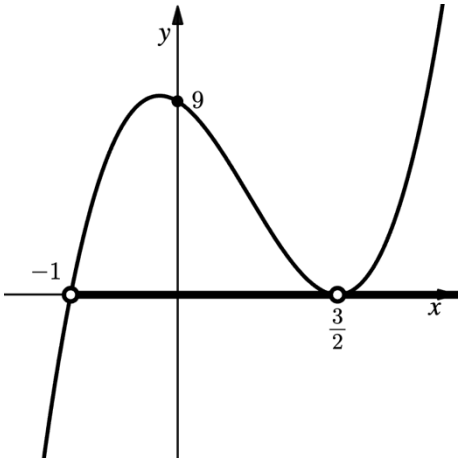
$(2x-1)(x^2+9x+20)>0$

$\Rightarrow (2x-1)(x+4)(x+5)>0$



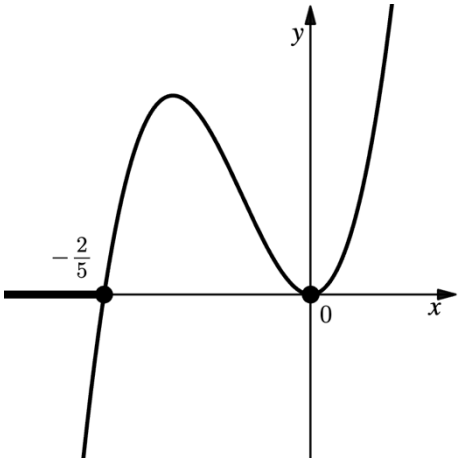
$\{x: x \in \mathbb{R}, -5 < x < -4\} \cup \left\{x: x \in \mathbb{R}, x > \frac{1}{2}\right\}$

2) f)



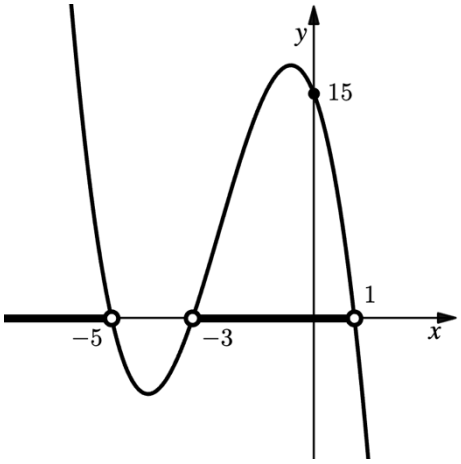
$\left\{x: x \in \mathbb{R}, -1 < x < \frac{3}{2}\right\} \cup \left\{x: x \in \mathbb{R}, x > \frac{3}{2}\right\}$

2) g)



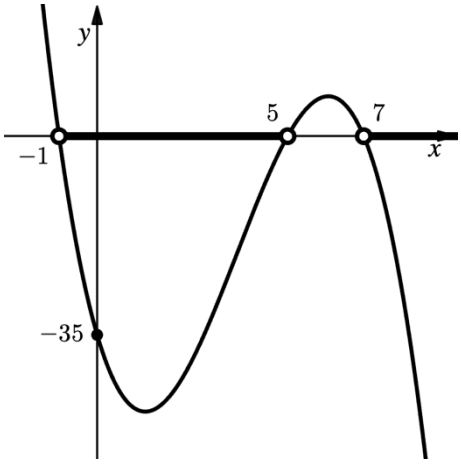
$\left\{x: x \in \mathbb{R}, x \leq -\frac{2}{5}\right\} \cup \{x: x \in \mathbb{R}, x = 0\}$

3) a)



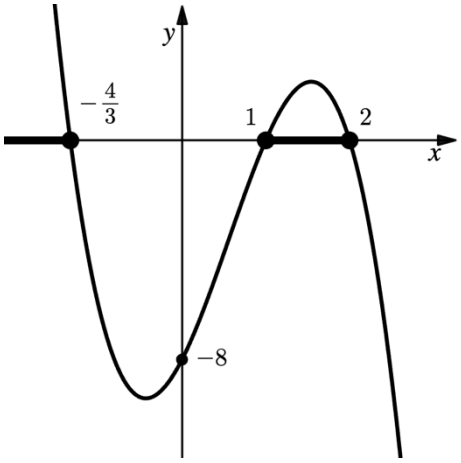
$\{x: x \in \mathbb{R}, x < -5\} \cup \{x: x \in \mathbb{R}, -3 < x < 1\}$

3) b)



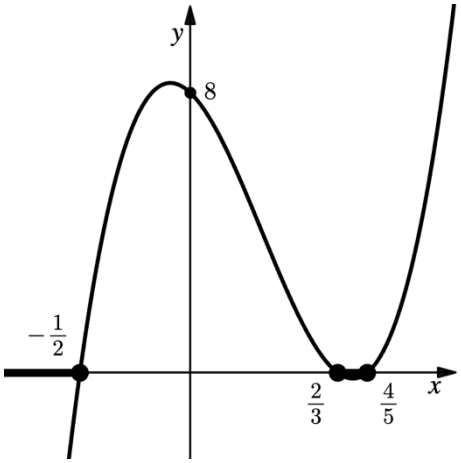
$\{x: x \in \mathbb{R}, -1 < x < 5\} \cup \{x: x \in \mathbb{R}, x > 7\}$

3) c)



$\left\{x: x \in \mathbb{R}, x \leq -\frac{4}{3}\right\} \cup \{x: x \in \mathbb{R}, 1 \leq x \leq 2\}$

3) d)

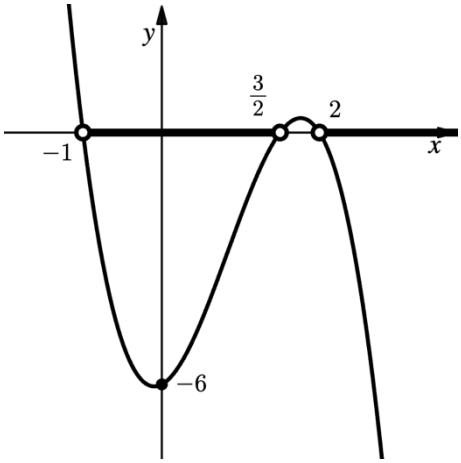


$\left\{x: x \in \mathbb{R}, x \leq -\frac{1}{2}\right\} \cup \left\{x: x \in \mathbb{R}, \frac{2}{3} \leq x \leq \frac{4}{5}\right\}$

3) e)

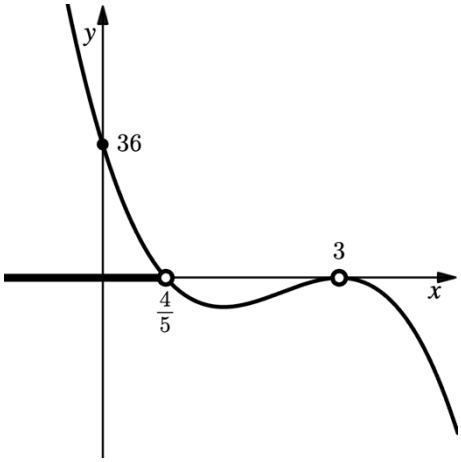
$(3-2x)(x^2-x-2)<0$

$\Rightarrow (3-2x)(x-2)(x+1)<0$



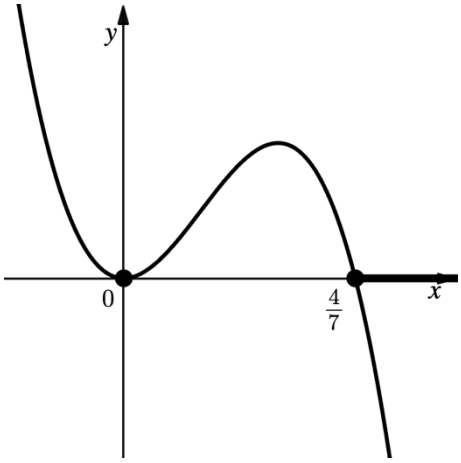
$\left\{x: x\in\mathbb{R}, -1<x<\frac{3}{2}\right\}\cup\{x: x\in\mathbb{R}, x>2\}$

3) f)



$\left\{x: x\in\mathbb{R}, x<\frac{4}{5}\right\}$

3) g)

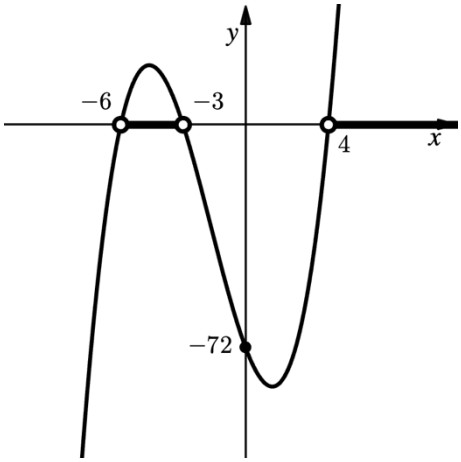


$\left\{x: x \in \mathbb{R}, x \geq \frac{4}{7}\right\}$

4) a)

$x^3 + 5x^2 - 18x - 72 > 0$

$\Rightarrow (x+3)(x-4)(x+6) > 0$

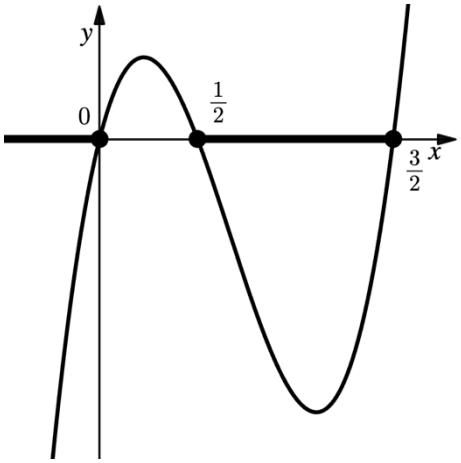


$x \in (-6, -3) \cup (4, \infty)$

4) b)

$4x^3 - 8x^2 + 3x \leq 0$

$\Rightarrow (2x-1)(2x-3)x \leq 0$

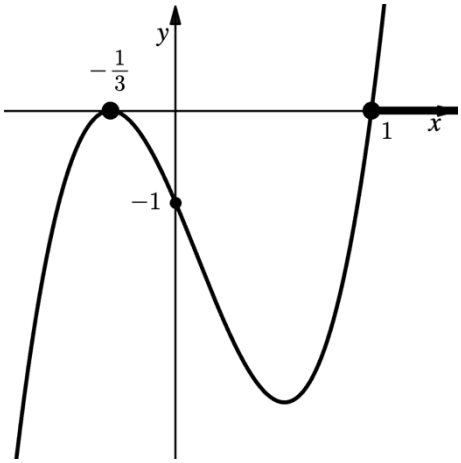


$x \in (-\infty, 0] \cup \left[\frac{1}{2}, \frac{3}{2}\right]$

4) c)

$9x^3 - 3x^2 - 5x - 1 \geq 0$

$\Rightarrow (3x+1)^2(x-1) \geq 0$

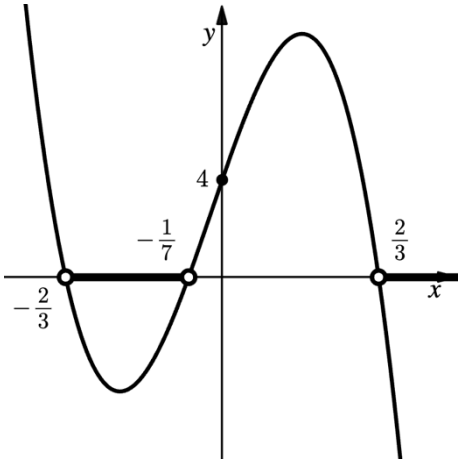


$x \in \left\{-\frac{1}{3}\right\} \cup [1, \infty)$ or $x \in \left[-\frac{1}{3}, -\frac{1}{3}\right] \cup [1, \infty)$

4) d)

$4 + 28x - 9x^2 - 63x^3 < 0$

$\Rightarrow (2 - 3x)(7x + 1)(3x + 2) < 0$

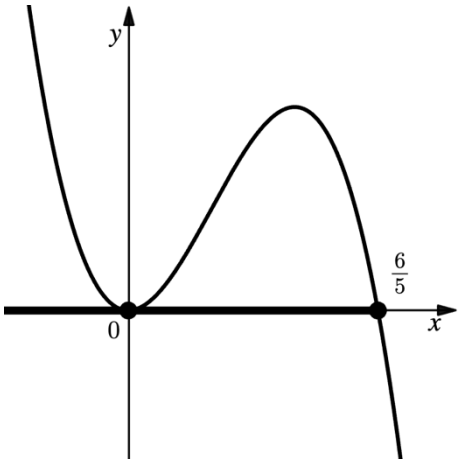


$x \in \left(-\frac{2}{3}, -\frac{1}{7}\right) \cup \left(\frac{2}{3}, \infty\right)$

4) e)

$6x^2 - 5x^3 \geq 0$

$\Rightarrow (6 - 5x)x^2 \geq 0$

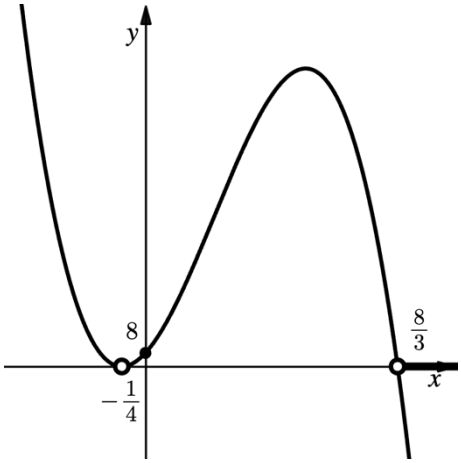


$x \in \left(-\infty, \frac{6}{5}\right]$

4) f)

$8 + 61x + 104x^2 - 48x^3 < 0$

$\Rightarrow (8 - 3x)(4x + 1)^2 < 0$



$x \in \left(\frac{8}{3}, \infty\right)$

5) a)

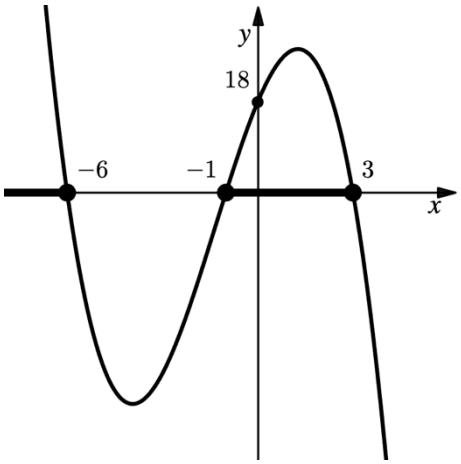
$(1 + x)(3 - x)(x + 5) \geq (1 + x)(x - 3)$

$\Rightarrow (1 + x)(3 - x)(x + 5) - (1 + x)(x - 3) \geq 0$

$\Rightarrow (1 + x)(3 - x)(x + 5) + (1 + x)(3 - x) \geq 0$

$\Rightarrow (1 + x)(3 - x)\big((x + 5) + 1\big) \geq 0$

$\Rightarrow (1 + x)(3 - x)(x + 6) \geq 0$



$\{x: x \in \mathbb{R}, x \leq -6\} \cup \{x: x \in \mathbb{R}, -1 \leq x \leq 3\}$

5) b)

$(2+x)(x-4)x \leq (4-x)x^2$

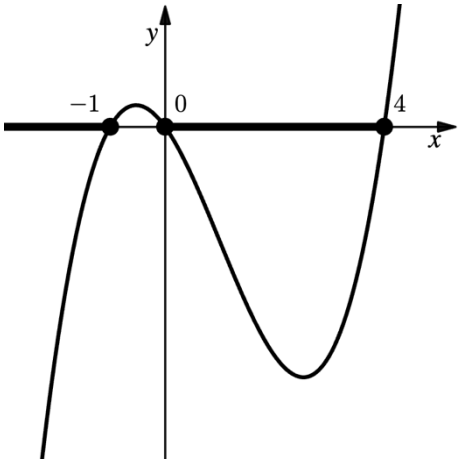
$\Rightarrow (2+x)(x-4)x - (4-x)x^2 \leq 0$

$\Rightarrow (2+x)(x-4)x + (x-4)x^2 \leq 0$

$\Rightarrow x(x-4)\big((2+x)+x\big) \leq 0$

$\Rightarrow x(x-4)(2+2x) \leq 0$

$\Rightarrow x(x-4)(1+x) \leq 0$



$\{x: x \in \mathbb{R}, x \leq -1\} \cup \{x: x \in \mathbb{R}, 0 \leq x \leq 4\}$

5) c)

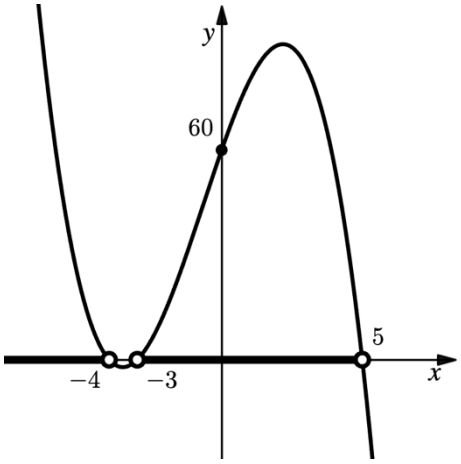
$(5-x)(3+x) > (3+x)^2(x-5)$

$\Rightarrow (5-x)(3+x) - (3+x)^2(x-5) > 0$

$\Rightarrow (5-x)(3+x) + (3+x)^2(5-x) > 0$

$\Rightarrow (5-x)(3+x)\big(1+(3+x)\big) > 0$

$\Rightarrow (5-x)(3+x)(4+x) > 0$



$\{x: x \in \mathbb{R}, x < -4\} \cup \{x: x \in \mathbb{R}, -3 < x < 5\}$

5) d)

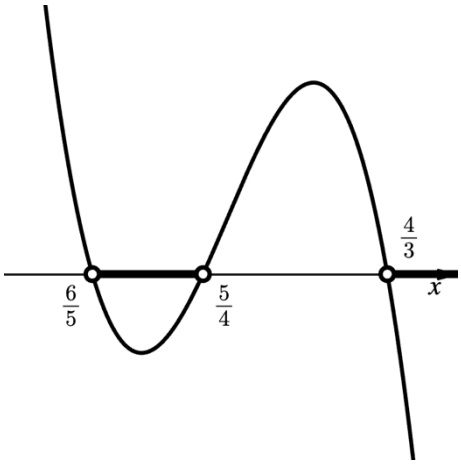
$(4-3x)^2(6-5x) < (3x-4)(5x-6)^2$

$\Rightarrow (3x-4)^2(6-5x) - (3x-4)(6-5x)^2 < 0$

$\Rightarrow (3x-4)(6-5x)\big((3x-4)-(6-5x)\big) < 0$

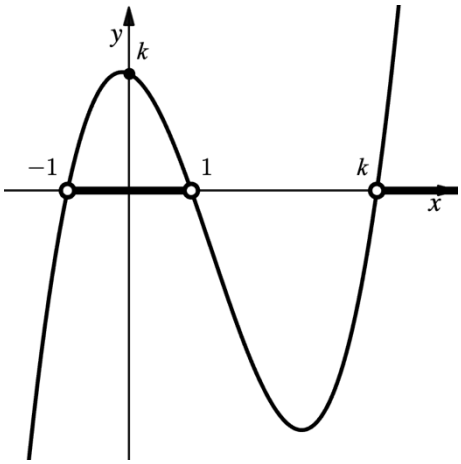
$\Rightarrow (3x-4)(6-5x)(8x-10) < 0$

$\Rightarrow (3x-4)(6-5x)(4x-5) < 0$



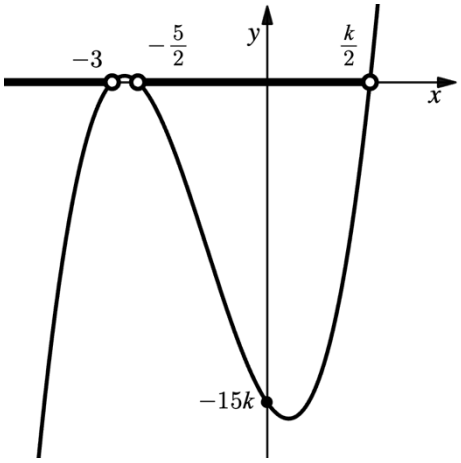
$\left\{x: x \in \mathbb{R}, \frac{6}{5} < x < \frac{5}{4}\right\} \cup \left\{x: x \in \mathbb{R}, x > \frac{4}{3}\right\}$

6) a)



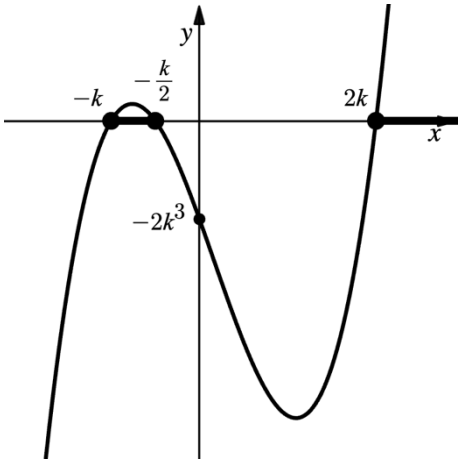
$\{x: x \in \mathbb{R}, -1 < x < 1\} \cup \{x: x \in \mathbb{R}, x > k\}$

6) b)



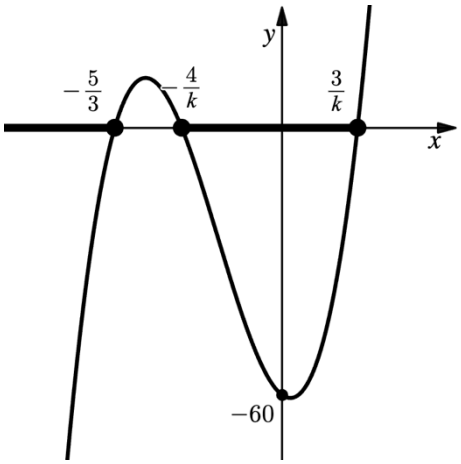
$\{x: x \in \mathbb{R}, x < -3\} \cup \left\{x: x \in \mathbb{R}, -\frac{5}{2} < x < \frac{k}{2}\right\}$

6) c)



$\left\{x: x \in \mathbb{R}, -k \leq x \leq -\frac{k}{2}\right\} \cup \{x: x \in \mathbb{R}, x \geq 2k\}$

6) d)



$\left\{x: x \in \mathbb{R}, x \leq -\frac{5}{3}\right\} \cup \left\{x: x \in \mathbb{R}, -\frac{4}{k} \leq x \leq \frac{3}{k}\right\}$

7)

Critical values are $x = -\frac{7}{2}$, $x = -2$ and $x = q$

$\therefore (2x + 7)(x + 2)(x - q) \geq 0$

$\Rightarrow (2x^2 - 2qx + 7x - 7q)(x + 2) \geq 0$

$\Rightarrow (2x^2 + (7 - 2q)x - 7q)(x + 2) \geq 0$

$7 - 2q = 19 \Rightarrow q = -6$

$p = -7(-6) = 42$

8)

Critical values are $x = -4$, $x = -\frac{9}{4}$ and $x = r$

$\therefore (x + 4)(4x + 9)(x - r) \leq 0$

$\Rightarrow (x + 4)(4x^2 - 4rx + 9x - 9r) \leq 0$

$\Rightarrow (x + 4)(4x^2 + (9 - 4r)x - 9r) \leq 0$

$\Rightarrow 4x^3 + (9 - 4r)x^2 - 9rx + 16x^2 + 4(9 - 4r)x - 36r \leq 0$

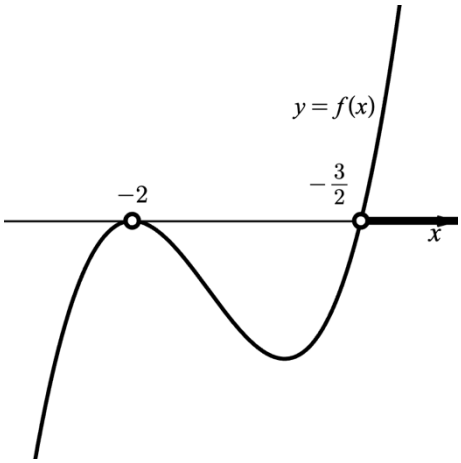
$\Rightarrow 4x^3 + (25 - 4r)x^2 + (36 - 25r)x - 36r \leq 0$

$25 - 4r = 17 \Rightarrow r = 2$

$q = 36 - 25(2) = -14$

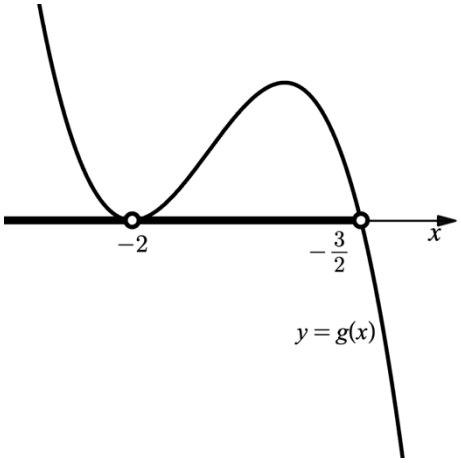
So $p = 4$, $q = -14$ and $r = 2$

9) a)



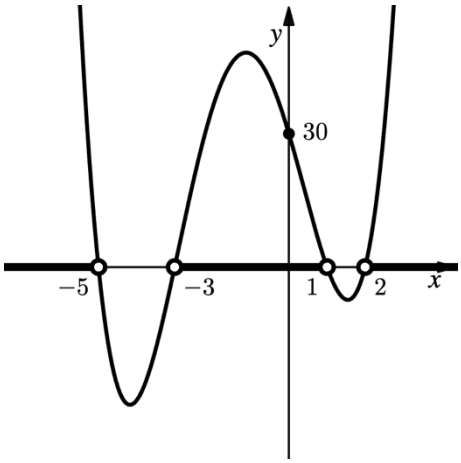
$x \in \left(-\frac{3}{2}, \infty\right)$

9) b)



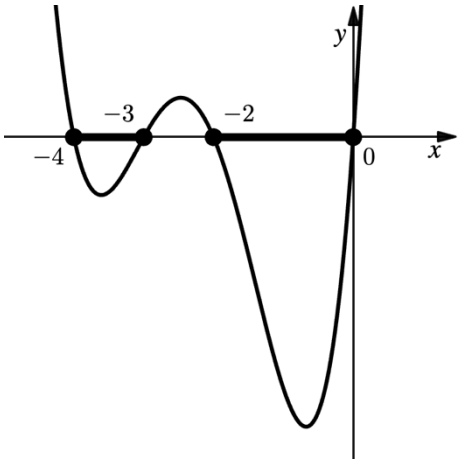
$x \in (-\infty, -2) \cup \left(-2, -\frac{3}{2}\right)$

10) a)



$\{x: x \in \mathbb{R}, x < -5\} \cup \{x: x \in \mathbb{R}, -3 < x < 1\} \cup \{x: x \in \mathbb{R}, x > 2\}$

10) b)

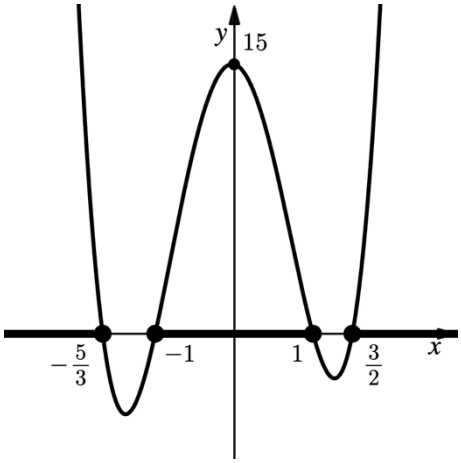


$\{x: x \in \mathbb{R}, -4 \leq x \leq -3\} \cup \{x: x \in \mathbb{R}, -2 \leq x \leq 0\}$

10) c)

$(2x-3)(3x+5)(x^2-1) \geq 0$

$\Rightarrow (2x-3)(3x+5)(x-1)(x+1) \geq 0$

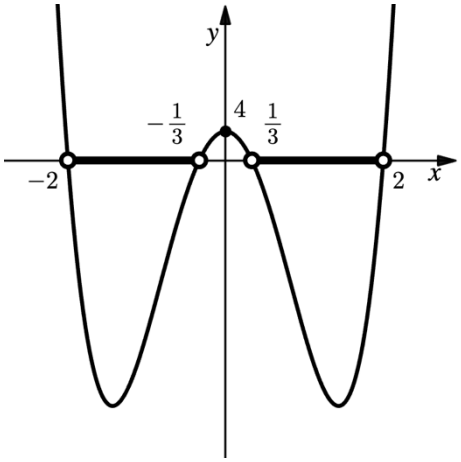


$\left\{x: x \in \mathbb{R}, x \leq -\frac{5}{3}\right\} \cup \{x: x \in \mathbb{R}, -1 \leq x \leq 1\} \cup \left\{x: x \in \mathbb{R}, x \geq \frac{3}{2}\right\}$

10) d)

$(x^2-4)(9x^2-1) < 0$

$\Rightarrow (x-2)(x+2)(3x-1)(3x+1) < 0$

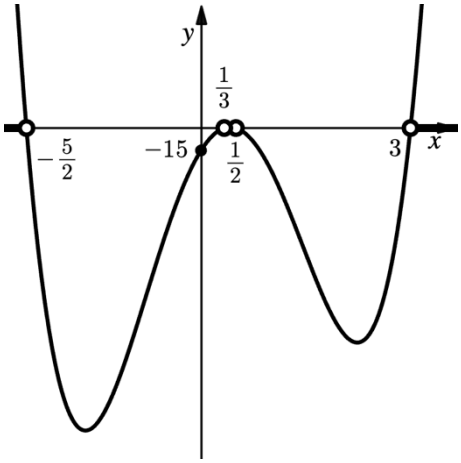


$\left\{x: x \in \mathbb{R}, -2 < x < -\frac{1}{3}\right\} \cup \left\{x: x \in \mathbb{R}, \frac{1}{3} < x < 2\right\}$

10) e)

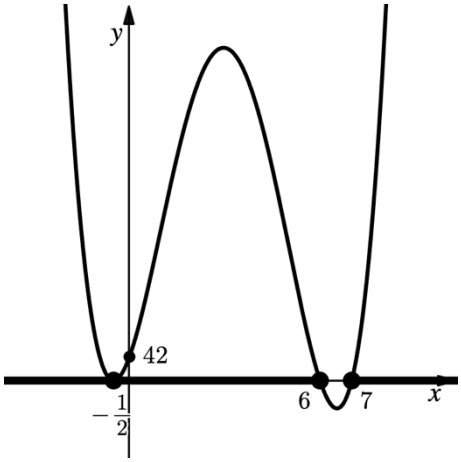
$(4x^2 + 8x - 5)(3x^2 - 10x + 3) > 0$

$\Rightarrow (2x - 1)(2x + 5)(3x - 1)(x - 3) > 0$



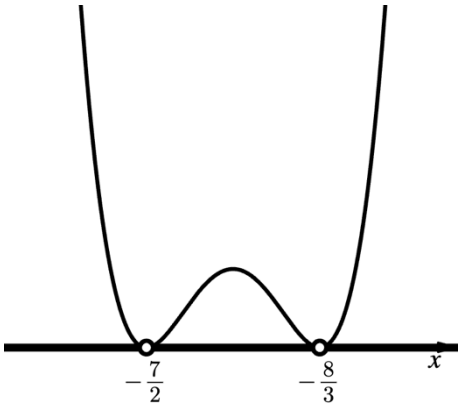
$\left\{x: x \in \mathbb{R}, x < -\frac{5}{2}\right\} \cup \left\{x: x \in \mathbb{R}, \frac{1}{3} < x < \frac{1}{2}\right\} \cup \{x: x \in \mathbb{R}, x > 3\}$

10) f)



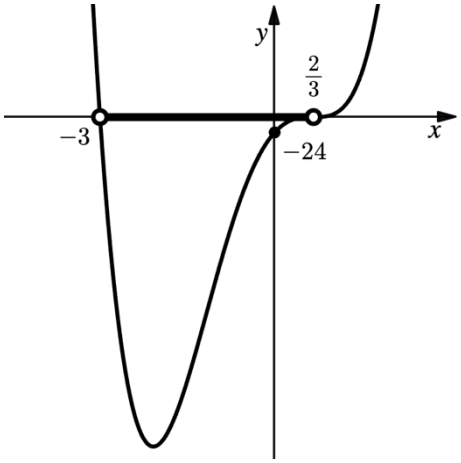
$\{x: x \in \mathbb{R}, x \leq 6\} \cup \{x: x \in \mathbb{R}, x \geq 7\}$

10) g)



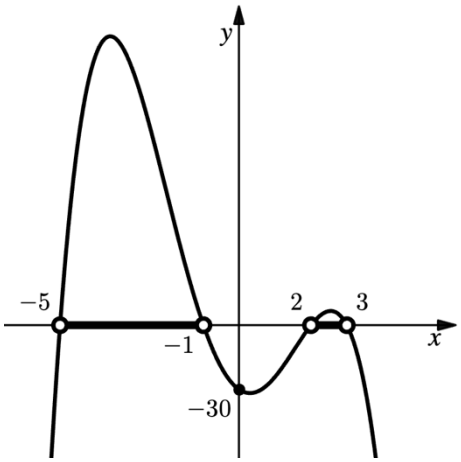
$\left\{x: x \in \mathbb{R}, x < -\frac{7}{2}\right\} \cup \left\{x: x \in \mathbb{R}, -\frac{7}{2} < x < -\frac{8}{3}\right\} \cup \left\{x: x \in \mathbb{R}, x > -\frac{8}{3}\right\}$

10) h)



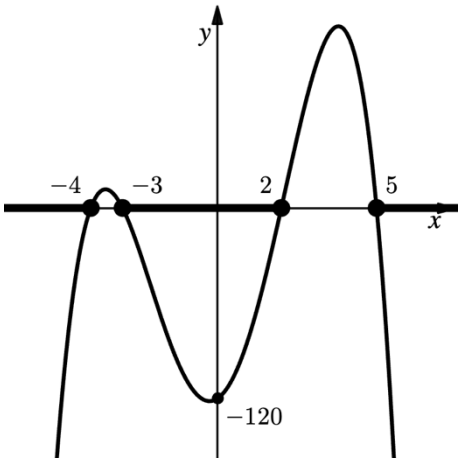
$\left\{x: x \in \mathbb{R}, -3 < x < \frac{2}{3}\right\}$

11) a)



$\{x: x \in \mathbb{R}, -5 < x < -1\} \cup \{x: x \in \mathbb{R}, 2 < x < 3\}$

11) b)

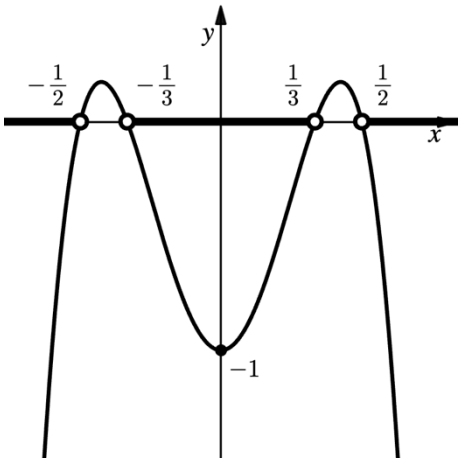


$\{x: x \in \mathbb{R}, x \leq -4\} \cup \{x: x \in \mathbb{R}, -3 \leq x \leq 2\} \cup \{x: x \in \mathbb{R}, x \geq 5\}$

11) c)

$(4x^2 - 1)(1 - 9x^2) < 0$

$\Rightarrow (2x - 1)(2x + 1)(1 - 3x)(1 + 3x) < 0$

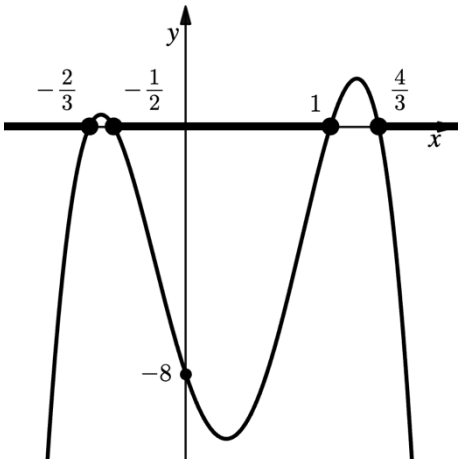


$\left\{x: x \in \mathbb{R}, x < -\frac{1}{2}\right\} \cup \left\{x: x \in \mathbb{R}, -\frac{1}{3} < x < \frac{1}{3}\right\} \cup \left\{x: x \in \mathbb{R}, x > \frac{1}{2}\right\}$

11) d)

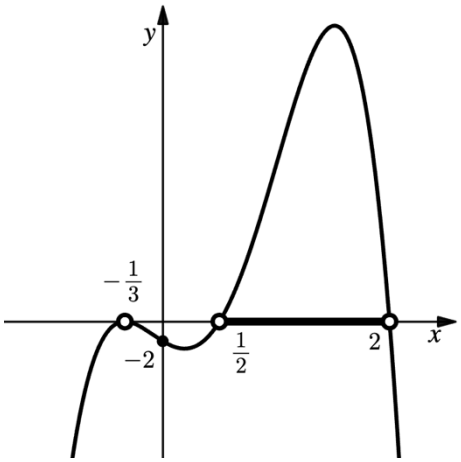
$(1 + x - 2x^2)(9x^2 - 6x - 8) \leq 0$

$\Rightarrow (1 - x)(2x + 1)(3x + 2)(3x - 4) \leq 0$



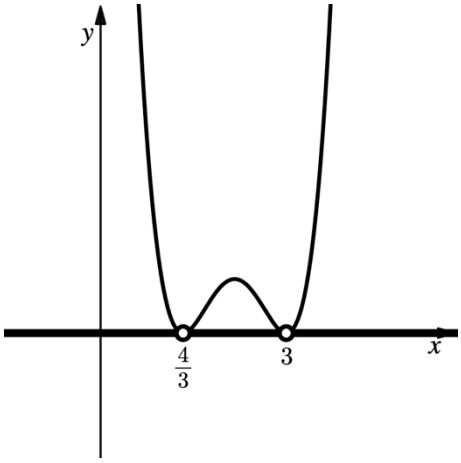
$\left\{x: x \in \mathbb{R}, x \leq -\frac{2}{3}\right\} \cup \left\{x: x \in \mathbb{R}, -\frac{1}{2} \leq x \leq 1\right\} \cup \left\{x: x \in \mathbb{R}, x \geq \frac{4}{3}\right\}$

11) e)



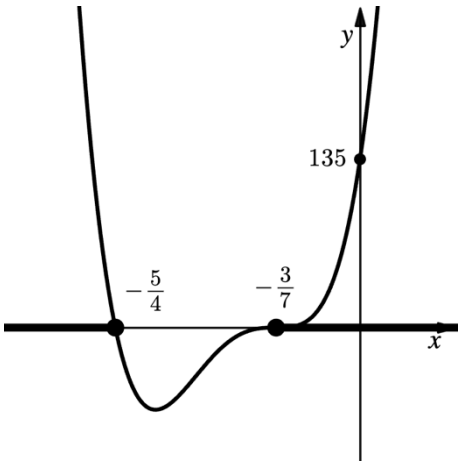
$\left\{x: x \in \mathbb{R}, \frac{1}{2} < x < 2\right\}$

11) f)



$\left\{x: x \in \mathbb{R}, x < \frac{4}{3}\right\} \cup \left\{x: x \in \mathbb{R}, \frac{4}{3} < x < 3\right\} \cup \{x: x \in \mathbb{R}, x > 3\}$

11) g)

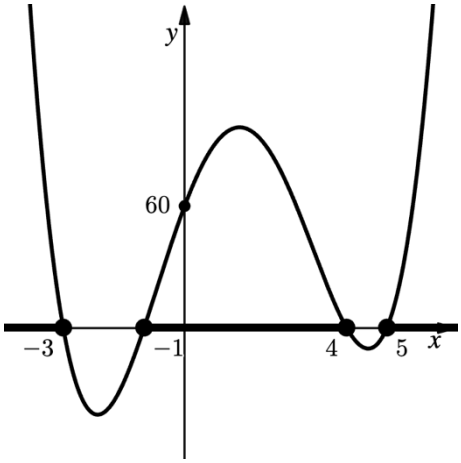


$\left\{x: x \in \mathbb{R}, x < -\frac{5}{4}\right\} \cup \left\{x: x \in \mathbb{R}, x > -\frac{3}{7}\right\}$

12) a)

$x^4 - 5x^3 - 13x^2 + 53x + 60 \geq 0$

$\Rightarrow (x+1)(x+3)(x-4)(x-5) \geq 0$

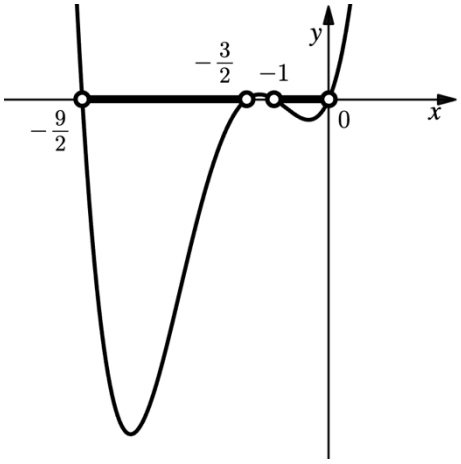


$x \in (-\infty, 3] \cup [-1, 4] \cup [5, \infty)$

12) b)

$4x^4 + 28x^3 + 51x^2 + 27x < 0$

$\Rightarrow (2x+3)(2x+9)(x+1)x < 0$

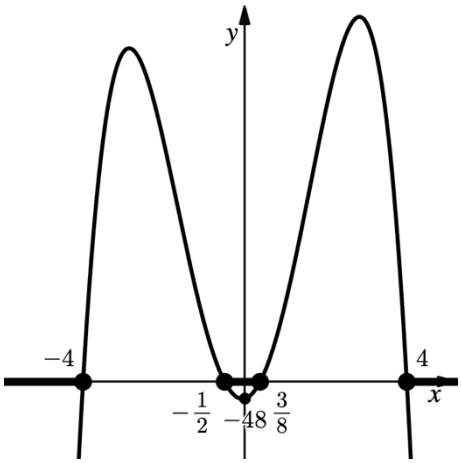


$x \in \left(-\frac{9}{2}, -\frac{3}{2}\right) \cup (-1, 0)$

12) c)

$259x^2 + 32x - 48 - 2x^3 - 16x^4 \leq 0$

$\Rightarrow (3 - 8x)(2x + 1)(x - 4)(x + 4) \leq 0$

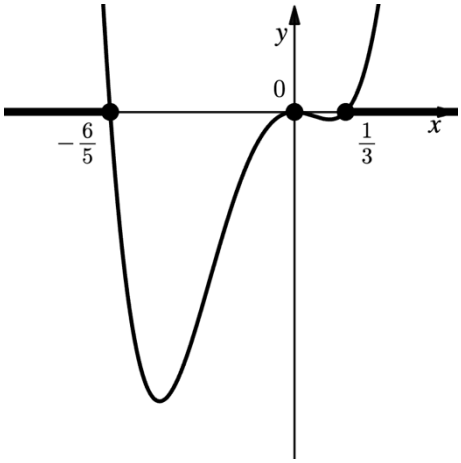


$x \in (-\infty, -4] \cup \left[-\frac{1}{2}, \frac{3}{8}\right] \cup [4, \infty)$

12) d)

$15x^4 + 13x^3 - 6x^2 \geq 0$

$\Rightarrow (5x + 6)(3x - 1)x^2 \geq 0$

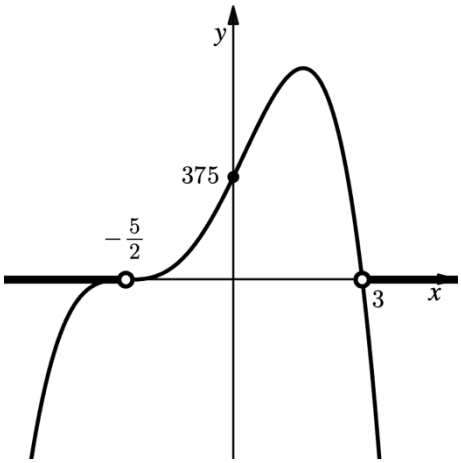


$x \in \left(-\infty, -\frac{6}{5}\right] \cup \{0\} \cup \left[\frac{1}{3}, \infty\right)$ or $x \in \left(-\infty, -\frac{6}{5}\right] \cup [0, 0] \cup \left[\frac{1}{3}, \infty\right)$

12) e)

$30x^2 + 325x + 375 - 36x^3 - 8x^4 < 0$

$\Rightarrow (2x + 5)^3(3 - x) < 0$



$x \in \left(-\infty, -\frac{5}{2}\right) \cup (3, \infty)$

13) a)

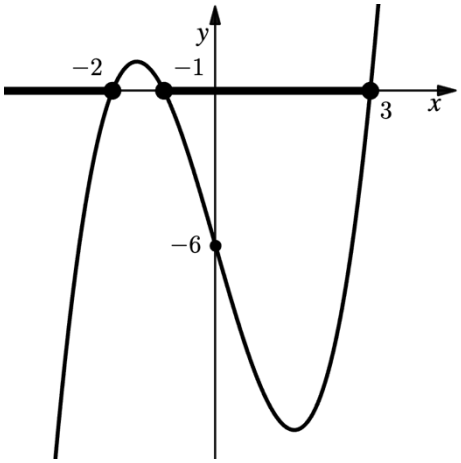
$(x + 2)(x - 3)(x + 1)(x - 5) \geq (x + 2)^2(x - 3)(x + 1)$

$\Rightarrow (x + 2)(x - 3)(x + 1)(x - 5) - (x + 2)^2(x - 3)(x + 1) \geq 0$

$\Rightarrow (x + 2)(x - 3)(x + 1)\big((x - 5) - (x + 2)\big) \geq 0$

$\Rightarrow (x + 2)(x - 3)(x + 1) \times -7 \geq 0$

$\Rightarrow (x + 2)(x - 3)(x + 1) \leq 0$



$\{x: x \in \mathbb{R}, x \leq -2\} \cup \{x: x \in \mathbb{R}, -1 \leq x \leq 3\}$

13) b)

$(5+x)(x+6)(3-x)x \leq (x-3)^2x^2$

$\Rightarrow (5+x)(x+6)(3-x)x - (x-3)^2x^2 \leq 0$

$\Rightarrow -(5+x)(x+6)(x-3)x - (x-3)^2x^2 \leq 0$

$\Rightarrow (5+x)(x+6)(x-3)x + (x-3)^2x^2 \geq 0$

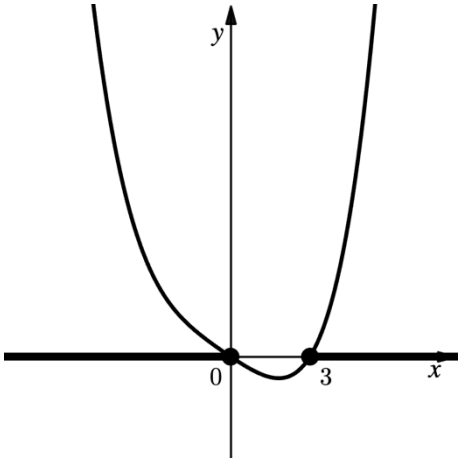
$\Rightarrow x(x-3)\big((5+x)(x+6) + (x-3)x\big) \geq 0$

$\Rightarrow x(x-3)\big((x^2+11x+30) + (x^2-3x)\big) \geq 0$

$\Rightarrow x(x-3)(2x^2+8x+30) \geq 0$

$\Rightarrow x(x-3)(x^2+4x+15) \geq 0$

For $x^2+4x+15=0$, $b^2-4ac = 4^2 - 4\times1\times15 = -44 < 0$, so there are no real roots



$\{x: x \in \mathbb{R}, x \leq 0\} \cup \{x: x \in \mathbb{R}, x \geq 3\}$

13) c)

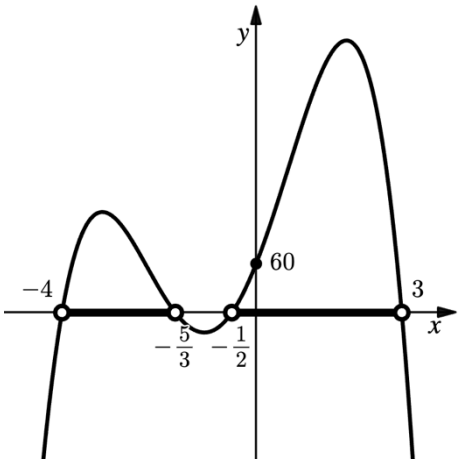
$(3-x)(2x+1)(4+x)^2 > (x-3)(2x+1)^2(4+x)$

$\Rightarrow (3-x)(2x+1)(4+x)^2 - (x-3)(2x+1)^2(4+x) > 0$

$\Rightarrow (3-x)(2x+1)(4+x)^2 + (3-x)(2x+1)^2(4+x) > 0$

$\Rightarrow (3-x)(2x+1)(4+x)\big((4+x)+(2x+1)\big) > 0$

$\Rightarrow (3-x)(2x+1)(4+x)(3x+5) > 0$



$\left\{x: x \in \mathbb{R}, -4 < x < -\frac{5}{3}\right\} \cup \left\{x: x \in \mathbb{R}, -\frac{1}{2} < x < 3\right\}$

13) d)

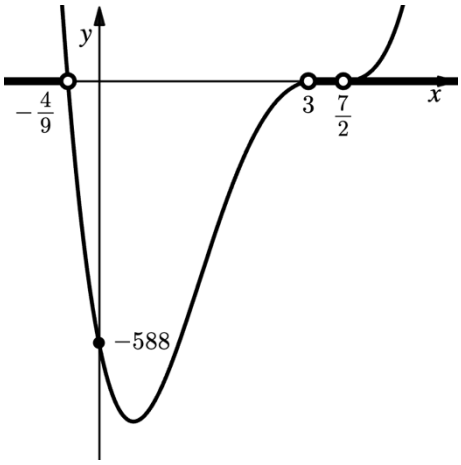
$(7-2x)^3(4+9x) < (2x-7)^2(4+9x)$

$\Rightarrow (7-2x)^3(4+9x) - (7-2x)^2(4+9x) < 0$

$\Rightarrow (7-2x)^2(4+9x)\big((7-2x)-1\big) < 0$

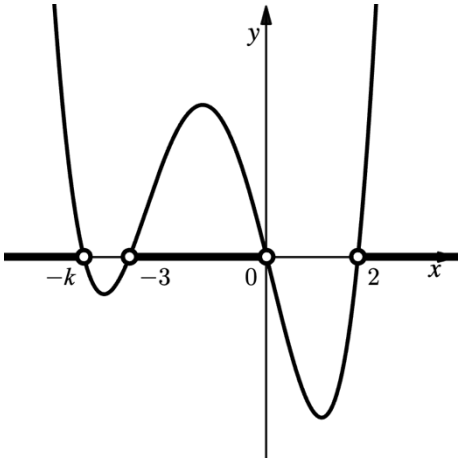
$\Rightarrow (7-2x)^2(4+9x)(6-2x) < 0$

$\Rightarrow (2x-7)^2(4+9x)(x-3) > 0$



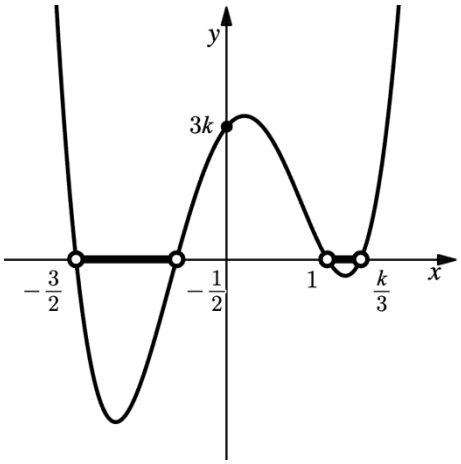
$\left\{x: x \in \mathbb{R}, x < -\frac{4}{9}\right\} \cup \left\{x: x \in \mathbb{R}, 3 < x < \frac{7}{2}\right\} \cup \left\{x: x \in \mathbb{R}, x > \frac{7}{2}\right\}$

14) a)



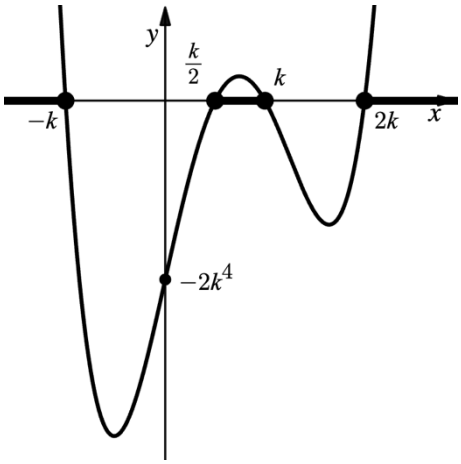
$\{x: x \in \mathbb{R}, x < -k\} \cup \{x: x \in \mathbb{R}, -3 < x < 0\} \cup \{x: x \in \mathbb{R}, x > 2\}$

14) b)



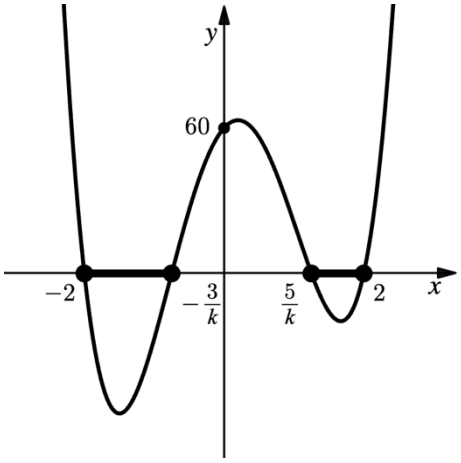
$\left\{x: x \in \mathbb{R}, -\frac{3}{2} < x < -\frac{1}{2}\right\} \cup \left\{x: x \in \mathbb{R}, 1 < x < \frac{k}{3}\right\}$

14) c)



$\{x: x \in \mathbb{R}, x \leq -k\} \cup \left\{x: x \in \mathbb{R}, \frac{k}{2} \leq x \leq k\right\} \cup \{x: x \in \mathbb{R}, x \geq 2k\}$

14) d)



$\left\{x: x \in \mathbb{R}, -2 \leq x \leq -\frac{3}{k}\right\} \cup \left\{x: x \in \mathbb{R}, \frac{5}{k} \leq x \leq 2\right\}$

15)

Critical values are $x = -\frac{3}{2}$, $x = -\frac{2}{5}$, $x = r$ and $x = \frac{3}{4}$

$\therefore (2x+3)(5x+2)(x-r)(4x-3) < 0$

$\Rightarrow (5x+2)(x-r)(2x+3)(4x-3) < 0$

$\Rightarrow (5x^2 - 5rx + 2x - 2r)(8x^2 + 6x - 9) < 0$

$\Rightarrow (5x^2 + (2-5r)x - 2r)(8x^2 + 6x - 9) < 0$

$p = 5$

$2 - 5r = -4 \Rightarrow r = \frac{6}{5}$

$q = -2\left(\frac{6}{5}\right) = -\frac{12}{5}$

So $p = 5$, $q = -\frac{12}{5}$ and $r = \frac{6}{5}$

16)

Critical values are $x = -\frac{5}{2}$, $x = -\frac{3}{2}$, $x = r$ and $x = 4$

$\therefore (2x+5)(2x+3)(x-r)(x-4) < 0$

$\Rightarrow (4x^2+6x+10x+15)(x^2-4x-rx+4r) < 0$

$\Rightarrow (4x^2+16x+15)(x^2-(4+r)x+4r) < 0$

$\Rightarrow 4x^4-(16+4r)x^3+16rx^2+16x^3-(64+16r)x^2+64rx+15x^2-(60+15r)x+60r < 0$

$\Rightarrow 4x^4-4rx^3-49x^2+(49r-60)x+60r < 0$

$p = 4$

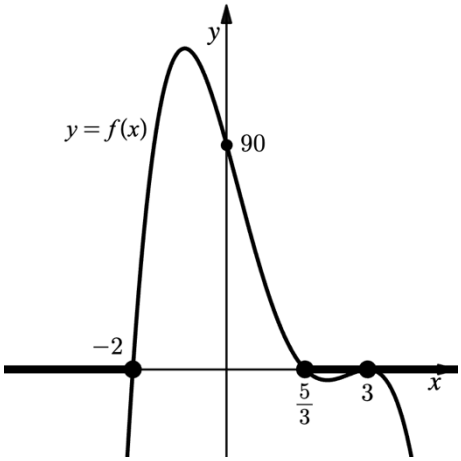
$-4r = -12 \Rightarrow r = 3$

$49(3)-60 = 87$ as required

$q = 60(3) = 180$

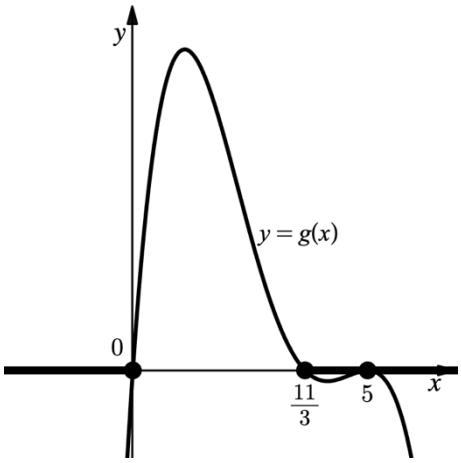
So $p = 4$, $q = 180$ and $r = 3$

17) a)



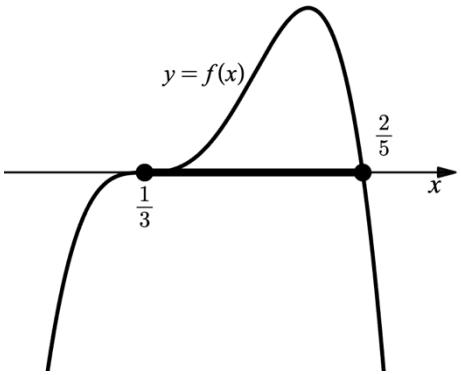
$x \in (-\infty, -2] \cup \left[\frac{5}{3}, \infty\right)$

17) b)



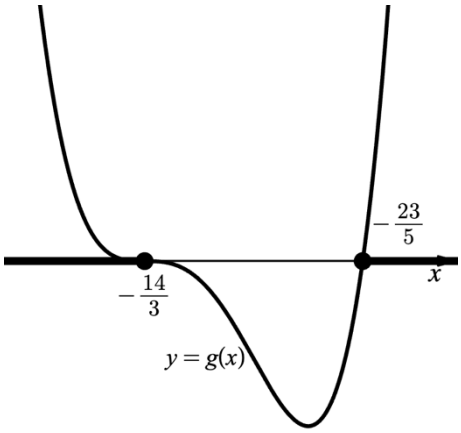
$x \in (-\infty, 0] \cup \left[\frac{11}{3}, \infty\right)$

18) a)



$x \in \left[\frac{1}{3}, \frac{2}{5}\right]$

18) b)



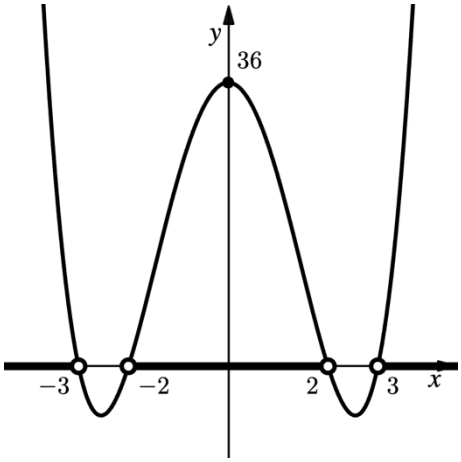
$x \in \left(-\infty, -\frac{14}{3}\right] \cup \left[-\frac{23}{5}, \infty\right)$

19) a)

$x^4 - 13x^2 + 36 > 0$

$\Rightarrow (x^2 - 4)(x^2 - 9) > 0$

$\Rightarrow (x - 2)(x + 2)(x - 3)(x + 3) > 0$



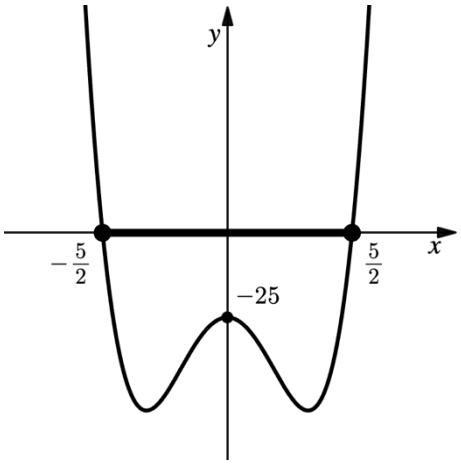
$\{x: x \in \mathbb{R}, x < -3\} \cup \{x: x \in \mathbb{R}, -2 < x < 2\} \cup \{x: x \in \mathbb{R}, x > 3\}$

19) b)

$4x^4 - 21x^2 - 25 \leq 0$

$\Rightarrow (x^2 + 1)(4x^2 - 25) \leq 0$

$\Rightarrow (x^2 + 1)(2x - 5)(2x + 5) \leq 0$



$\left\{x: x \in \mathbb{R}, -\frac{5}{2} < x < \frac{5}{2}\right\}$

20) a)

$e^{3x} - 12e^{2x} + 41e^{x} - 30 \leq 0$

$\Rightarrow (e^{x} - 1)(e^{x} - 5)(e^{x} - 6) \leq 0$

$e^{x} - 1 = 0 \Rightarrow e^{x} = 1 \Rightarrow x = 0$

$e^{x} - 5 = 0 \Rightarrow e^{x} = 5 \Rightarrow x = \ln 5$

$e^{x} - 6 = 0 \Rightarrow e^{x} = 6 \Rightarrow x = \ln 6$

Critical values at $x = 0$, $x = \ln 5$ and $x = \ln 6$

$x < 0$	$0 < x < \ln 5$	$\ln 5 < x < \ln 6$	$x > \ln 6$
$-$	$+$	$-$	$+$

$\{x: x \in \mathbb{R}, x \leq 0\} \cup \{x: x \in \mathbb{R}, \ln 5 \leq x \leq \ln 6\}$

20) b)

$e^{6x} - 6e^{4x} - e^{2x} + 30 > 0$

$\Rightarrow (e^{2x} + 2)(e^{2x} - 3)(e^{2x} - 5) > 0$

$e^{2x} + 2 = 0$ has no real solutions as $e^{2x} > 0$ for all $x \in \mathbb{R}$

$e^{2x} - 3 = 0 \Rightarrow e^{2x} = 3 \Rightarrow 2x = \ln 3 \Rightarrow x = \frac{1}{2}\ln 3$

$e^{2x} - 5 = 0 \Rightarrow e^{2x} = 5 \Rightarrow 2x = \ln 5 \Rightarrow x = \frac{1}{2}\ln 5$

Critical values at $x = \frac{1}{2}\ln 3$ and $x = \frac{1}{2}\ln 5$

$x < \frac{1}{2}\ln 3$	$\frac{1}{2}\ln 3 < x < \frac{1}{2}\ln 5$	$x > \frac{1}{2}\ln 5$
$+$	$-$	$+$

$\left\{x: x \in \mathbb{R}, x < \frac{1}{2}\ln 3\right\} \cup \left\{x: x \in \mathbb{R}, x > \frac{1}{2}\ln 5\right\}$

20) c)

$e^{4x} - 4e^{3x} - 25e^{2x} + 88e^{x} - 60 \geq 0$

$\Rightarrow (e^{x} - 1)(e^{x} - 2)(e^{x} + 5)(e^{x} - 6) \geq 0$

$e^{x} - 1 = 0 \Rightarrow e^{x} = 1 \Rightarrow x = 0$

$e^{x} - 2 = 0 \Rightarrow e^{x} = 2 \Rightarrow x = \ln 2$

$e^{x} + 5 = 0$ has no real solutions as $e^{x} > 0$ for all $x \in \mathbb{R}$

$e^{x} - 6 = 0 \Rightarrow e^{x} = 6 \Rightarrow x = \ln 6$

Critical values at $x = 0$, $x = \ln 2$ and $x = \ln 6$

$x < 0$	$0 < x < \ln 2$	$\ln 2 < x < \ln 6$	$x > \ln 6$
$-$	$+$	$-$	$+$

$\{x: x \in \mathbb{R}, 0 \leq x \leq \ln 2\} \cup \{x: x \in \mathbb{R}, x \geq \ln 6\}$

20) d)

$4e^{8x} - 8e^{6x} - 59e^{4x} + 63e^{2x} + 180 < 0$

$\Rightarrow (2e^{2x} + 3)(2e^{2x} - 5)(e^{2x} + 3)(e^{2x} - 4) < 0$

$2e^{2x} + 3 = 0$ has no real solutions as $e^{2x} > 0$ for all $x \in \mathbb{R}$

$2e^{2x} - 5 = 0 \Rightarrow e^{2x} = \frac{5}{2} \Rightarrow 2x = \ln\frac{5}{2} \Rightarrow x = \frac{1}{2}\ln\frac{5}{2}$

$e^{2x} + 3 = 0$ has no real solutions as $e^{2x} > 0$ for all $x \in \mathbb{R}$

$e^{2x} - 4 = 0 \Rightarrow e^{2x} = 4 \Rightarrow 2x = \ln 4 \Rightarrow x = \frac{1}{2}\ln 4 = \ln 2$

Critical values at $x = \frac{1}{2}\ln\frac{5}{2}$ and $x = \ln 2$

$x < \frac{1}{2}\ln\frac{5}{2}$	$\frac{1}{2}\ln\frac{5}{2} < x < \ln 2$	$x > \ln 2$
$+$	$-$	$+$

$\left\{x: x \in \mathbb{R}, \frac{1}{2}\ln\frac{5}{2} < x < \ln 2\right\}$

21)

$2^{12x} - 10 \times 2^{9x} + 160 \times 2^{3x} - 256 \leq 0$

$\Rightarrow (2^{3x} + 4)(2^{3x} - 2)(2^{3x} - 4)(2^{3x} - 8) \leq 0$

$2^{3x} + 4 \neq 0$

$2^{3x} - 2 = 0 \Rightarrow 2^{3x} = 2 \Rightarrow 3x = 1 \Rightarrow x = \frac{1}{3}$

$2^{3x} - 4 = 0 \Rightarrow 2^{3x} = 4 \Rightarrow 3x = 2 \Rightarrow x = \frac{2}{3}$

$2^{3x} - 8 = 0 \Rightarrow 2^{3x} = 8 \Rightarrow 3x = 3 \Rightarrow x = 1$

Critical values at $x = \frac{1}{3}$, $x = \frac{2}{3}$ and $x = 1$

$x < \frac{1}{3}$	$\frac{1}{3} < x < \frac{2}{3}$	$\frac{2}{3} < x < 1$	$x > 1$
$-$	$+$	$-$	$+$

$x \in \left(-\infty, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right]$

22)

$(2\sin x - 1)(\sin x + 3)(\sin x + k) > 0$

$2\sin x - 1 > 0 \Rightarrow \sin x > \frac{1}{2}$

Has critical values when $\sin x = \frac{1}{2} \Rightarrow x = \left\{\frac{\pi}{6}, \frac{5\pi}{6}\right\}$

So $\sin x > \frac{1}{2}$ when $\frac{\pi}{6} < x < \frac{5\pi}{6}$

Meanwhile, $\sin x + 3 > 0$ for all $x \in \mathbb{R}$

So we need $\sin x + k > 0$ for all $x \in \mathbb{R}$

Minimum value of $\sin x$ is -1 so $-1 + k > 0 \Rightarrow k > 1$

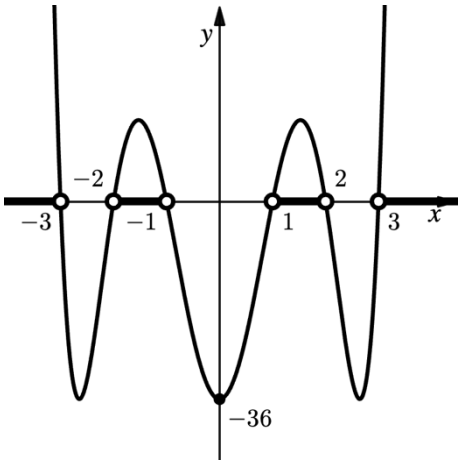
23) a)

$x^6 - 14x^4 + 49x^2 - 36 > 0$

$\Rightarrow (x^2 - 1)(x^2 - 4)(x^2 - 9) > 0$

$\Rightarrow (x - 1)(x + 1)(x - 2)(x + 2)(x - 3)(x + 3) > 0$

$x < -3$	$-3 < x < -2$	$-2 < x < -1$	$-1 < x < 1$	$1 < x < 2$	$2 < x < 3$	$x > 3$
$+$	$-$	$+$	$-$	$+$	$-$	$+$



$\{x: x \in \mathbb{R}, x < -3\} \cup \{x: x \in \mathbb{R}, -2 < x < -1\}$

$\cup \{x: x \in \mathbb{R}, 1 < x < 2\} \cup \{x: x \in \mathbb{R}, x > 3\}$

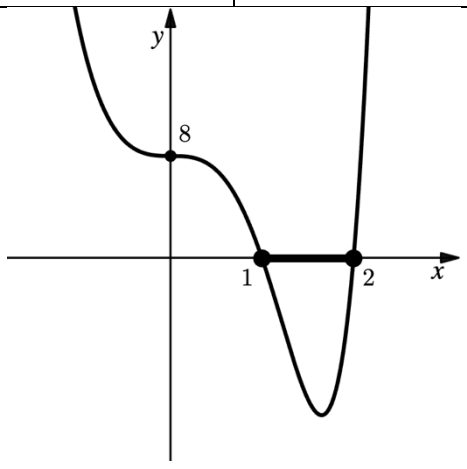
23) b)

$x^6 - 9x^3 + 8 \leq 0$

$\Rightarrow (x^3 - 1)(x^3 - 8) \leq 0$

Critical values at $x = 1$ and $x = 2$

$x < 1$	$1 < x < 2$	$x > 2$
$+$	$-$	$+$



$\{x: x \in \mathbb{R}, 1 \leq x \leq 2\}$

23) c)

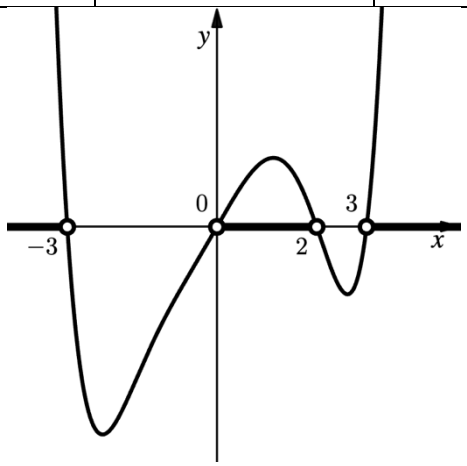
$x^6 - 9x^4 - 8x^3 + 72x > 0$

$\Rightarrow (x^2 + 2x + 4)(x - 3)(x + 3)(x - 2)x > 0$

$x^2 + 2x + 4 = 0$ has $b^2 - 4ac = 2^2 - 4 \times 1 \times 4 = -12 < 0$ so has no real roots.

$\therefore$ Critical values at $x = -3$, $x = 0$, $x = 2$ and $x = 3$

$x < -3$	$-3 < x < 0$	$0 < x < 2$	$2 < x < 3$	$x > 3$
$+$	$-$	$+$	$-$	$+$



$\{x: x \in \mathbb{R}, x < -3\} \cup \{x: x \in \mathbb{R}, 0 < x < 2\} \cup \{x: x \in \mathbb{R}, x > 3\}$